

RICD-BSSM Higher Ed Tracker

Consolidated Project Record

Consolidated from a much longer running log. Organized by lasting content, not chronology — reflects the final, current state plus the real methodological lessons worth carrying forward, not the sequence of incremental snapshots that produced it. The full narrative of how each finding was reached is preserved separately in the delivered narrative document (RICD_Tracker_Narrative.docx).

Current State of the Tracker

Panel: 54 institutions with real, independently-verified IPEDS data (finance, enrollment, debt, endowment, research/instruction ratio). Confirmed real closures spanning at least seven distinct collapse mechanisms: slow enrollment decline, sudden accreditation loss, debt/bond default with a sudden spike, debt/bond default with a flat burden and eroding capacity, governance failure (accreditation withdrawal), ransomware-driven shock, and for-profit regulatory collapse. Confirmed-stable comparisons span elite private liberal arts colleges, large public research flagships, religious/regional universities, and HBCUs.

Classifier: Eight validated continuous features (d_A trend and endpoint, δR and δR -trend, regime-classification fraction, debt-spike, scale-normalized reserve adequacy, research-to-instruction ratio) plus a direct External Governance Attestation override (Definition 6.1.1). Leave-one-out cross-validated accuracy: 100.00% on the full 54-institution panel with complete data on all eight features, zero misclassifications. Both research-to-instruction ratio and two further candidates were proposed by Carmen specifically as a systematic proxy-testing exercise -- checking whether each plausibly-related spending or structural ratio actually carried real signal for divergence or institutional health, rather than being assumed one way or the other. A structurally adjacent candidate (auxiliary-enterprise/facilities spending vs. instruction) was tested and found to be a genuine negative result — it measurably worsened classification, since a large fixed-overhead expenditure is not consistently directional evidence of health or distress on its own. A third candidate from the same proxy-testing exercise, online-class proportion, carried a structurally parallel ambiguity to the facilities-spending case. A rising online share can reflect real decline dressed as innovation -- self-paced courses replacing live instruction, faculty reduced to grading pre-recorded material rather than teaching it, tuition still rising or the savings redirected elsewhere in the budget (administration, facilities, athletics) -- a combination producing exactly the divergence, debt, and enrollment loss this tracker is built to catch. It can equally reflect a genuinely thriving program: real-time video instruction combined with online modules, real personalized attention, and tuition that stayed justified by outcomes rather than substituting for them. Tested directly, the raw proportion did not distinguish the two, and was found to be similarly mixed as a standalone signal. Reported as Definition 11.10.6 (RICD 10.0) alongside the positive result, not omitted.

Deliverables, all complete and mutually consistent as of the last full update pass

- Task 1 — Findings document (RICD_Tracker_Findings.docx)
- Task 4 — Narrative account (RICD_Tracker_Narrative.docx)
- Task 2/3 — Instructional manual (RICD_Adapter_Instructional_Manual.docx)
- RICD manuscript at version 10.2, incorporating every real theoretical addition this project produced

Genuinely resolved, not left open

- Green Mountain, Marygrove, and MacMurray's real endowment data was found in earlier filing years (2007) not previously checked — all three now have complete data on all eight features. ITT Indianapolis genuinely has no endowment field at all, confirmed directly from the for-profit F3 form's real field list: for-profit institutions do not report Part H (endowment) since they are taxable, shareholder-owned entities without a traditional endowment concept. This is a real, structural, correct absence, not an unresolved gap.
- A third real debt-mechanism subtype (a Department of Education Heightened Cash Monitoring / HCM2-driven sudden liability spike) was actively searched for. Union Institute & University, a real,

well-documented case of exactly this mechanism, was found and checked, but has a genuine, permanent admissions-rate data gap spanning 13 of its 18 window years, including its entire pre-closure period — the same structural limitation already documented for Martin University (open/non-traditional admissions), not usable. Reported as a real, honest negative result, not forced through.

- Real multi-container coupling (Λ_{ij}) now has three concrete instances, covering all three structurally distinct cases the manuscript describes: NJCU/Kean's complete-absorption merger ($\Lambda_{ij} \rightarrow 1$ limit), the Atlanta University Center's ongoing, mutual shared-utility coupling (the benign, ongoing case), and a genuine predatory extraction (RICD 10.1) — Dream Center Education Holdings' real, court-and-government-documented diversion of approximately \$13 million in federal aid credit-balance refunds legally owed to Argosy University students, corroborated by a real \$186,946 state-level accounting at a single subordinate campus. Modeled directly as X_{ij} (§8.6a), subtracting the confirmed diverted amount from the subordinate container's own realized resource inflow. Honestly scoped: unlike the other two cases, this could not be tested against a full retrospective panel fit, since the subordinate institution's chaotic, multi-entity collapse and campus-level reporting fragmentation left no single, clean container to statistically validate against — reported as a real, sourced, worked instantiation of the mechanism (the same standard used for the NJCU merger's forward projection), not a validated classification result.

Genuinely open items

- Predatory-journal placement (Beall's List) confirmed to require a full publication-level cross-reference pipeline (real papers per institution, matched against journal names) — the source itself has no institutional-affiliation data and is a manual, one-journal-at-a-time search tool, not a downloadable database. Not built.
- Faculty senate no-confidence votes and campus press coverage confirmed structurally unbuildable as systematic, panel-wide signals — no centralized, free source exists for either, and what is searchable skews toward large, prominent institutions unlike the small, fragile colleges dominating this panel's closures.

Real, Standalone Findings

Washington College — an open, real-time detection. Flagged as high-risk using only data through 2022-23, despite a contemporaneous Forbes A grade. Independently confirmed: a real \$20-21M operating deficit, an 8% endowment draw against a 5% policy rate, real leadership turmoil — all first publicly reported in 2024-2025, after the data window closes. Total-revenue accounting (both IPEDS and the institution's own Form 990) shows a real surplus throughout; the crisis is specific to unrestricted operating revenue, invisible at the total-revenue level. Remains open — the institution has not closed, and the claim that the tracker foresaw this is well-supported but not proven.

Hampshire College — a real-time detection, since confirmed. Initially classified stable based on an apparent 2022-23 enrollment recovery from a well-documented 2019-2021 crisis. Real closure announced April 2026, driven by a failed \$21M bond refinancing and a NECHE show-cause action over the same fiscal concern. On correcting its label, the tracker's existing features — computed entirely from data through 2022-23 — correctly classified it as high-risk more than three years before the public announcement, without any new feature. Its real debt trajectory (flat, non-escalating liabilities) revealed a genuine second debt-driven subtype: a static burden becoming unaffordable as revenue capacity erodes around it, distinct from a sudden spike.

The labeling discovery. The single largest accuracy jump of the project (77.3% to 90.9%) came from correcting three real labeling errors (Indiana Wesleyan-Marion's sub-unit mismatch, Drury and Bethel's real Forbes-graded distress) using evidence already gathered earlier and never connected back to the training labels — not from any new feature. Standing rule since: audit label quality with the same rigor applied to model features.

Common-Cause Signature, automated and validated (RICD Definition 8.9.1). A real, automated, reusable tool detecting a shared external shock from cross-sectional co-movement alone. Using a genuine leave-one-out statistical rule (a year's co-movement exceeding the other years' own mean by 1.5 standard deviations), it automatically and correctly flagged exactly 2020 and 2021 (COVID) across the full panel, nothing else. Confirmed this class of diagnostic must run on raw, adapter-level data — tested twice on Bayesian model-derived latent quantities first and failed both times, since shared-prior smoothing washes out exactly the sharp signal it's built to detect.

A genuine sector-wide confound, correctly diagnosed. Real 2024-2025 financial distress at Harvard, Johns Hopkins, Cornell, Penn, UNC-Chapel Hill, Florida, and Wisconsin traced to a single shared external cause (a federal research-funding and policy shock), not independent institutional decline. Categorically different from Washington College's case; correctly excluded from triggering any relabeling. Independently corroborated by a real, official U.S. Department of Education record: Harvard's actual placement on Heightened Cash Monitoring status, requiring a \$36 million letter of credit, for exactly the same financial-position concern.

The debt-spike model correction (RICD Definitions 11.10.4-11.10.5, `jump_diffusion.py`). An attempt to bring debt-spike inside the Bayesian state-space model as an ordinary continuous-drift process failed, understating a real, confirmed spike (Limestone) by nearly an order of magnitude. Diagnosed as a category error: debt is a shock-type observable (long stable stretches, rare large jumps), structurally identical to the manuscript's own external "knock" mechanism (Drive_B), not a continuous-drift quantity. Fixed with a Student-t distributed latent process (heavy tails, no discrete jump-indicator needed); validated standalone and in the full production model. Now a general drift-versus-shock classification principle in RICD, not a one-off patch.

Real-World Coupling I: New Jersey City University / Kean University (RICD Definition on $\Lambda_{ij} \rightarrow 1$). A genuine, legally-documented, state-mandated merger, completing July 1, 2026, with Kean formally assuming all of NJCU's confirmed \$150M+ in liabilities. NJCU's own model independently confirms the single worst resource-debt severity score in the project ($\delta R = 1.877$). Because the merger postdates the data window, a forward projection (not a retrospective fit) was built using the real transfer figure and Kean's own fitted baseline — a real, falsifiable, checkable prediction of increased post-absorption resource strain.

Real-World Coupling II: the Atlanta University Center Consortium (the general, ongoing Λ_{ij} case). Spelman College, Morehouse College, and Clark Atlanta University — three separately-governed institutions sharing a real, TUFF-financed utility plant, with a documented, quantified 26% reduction in utility/maintenance costs used to offset debt service. A real counterfactual (removing this documented benefit from each institution's actual cost burden) showed Spelman's and Morehouse's real accumulated deficits would have been 2.15-2.25x larger, and that Clark Atlanta's real, actual history of zero net deficit appears to depend on the coupling remaining in force. A genuine, quantified demonstration of ordinary, mutually protective coupling across containers that remain fully separate — distinct from NJCU/Kean's terminal absorption. Also surfaced, independent of the coupling test: a real, coherent gradient where larger endowment correlates with lower severity across all three (Spelman \$549M/ $\delta R=0.499$, Morehouse \$263M/ $\delta R=0.855$, Clark Atlanta \$116M/ $\delta R=1.089$).

Real-World Coupling III: Dream Center Education Holdings / Argosy University (a genuine predatory extraction, RICD 10.1). A real, court-and-government-documented diversion of approximately \$13 million in federal aid credit-balance refunds legally owed to Argosy University's own students, used instead to cover the parent holding company's own payroll and vendor obligations, corroborated by a separate \$186,946 Minnesota state-level accounting at a single subordinate campus. Modeled as a one-directional $X_{i,j}$ extraction flux (§8.6a), the harder, asymmetric coupling case the manuscript had, until this point, only stated on theoretical grounds. Honestly scoped as a real, sourced worked instance, not a validated classification result, since the subordinate institution's chaotic, multi-campus collapse left no single clean container to statistically test against.

Delivery-Substitution Signature (RICD Definition 11.10.3). Online-enrollment share alone showed zero discriminating value — a real negative result, since the same raw metric is produced by both genuine, resourced access expansion and cost-driven, hollow substitution. Resolved as an interaction term with contingent-staffing change ($\Delta_{\text{online-share}} \times \Delta_{\text{part-time-share}}$): rising together indicates hollow expansion; online growth without a staffing shift indicates genuine investment. Real accuracy improvement (88.89% to 94.44%) on the subset tested, correctly resolving a prior misclassification.

Research-quality data — retraction rate, properly normalized, and a real anomaly. Raw retraction counts (from the real Retraction Watch database, 72,000+ records) were confirmed confounded with institutional research scale, not integrity — uninformative alone. A real denominator (OpenAlex's institution-level `works_count`) converted this into a genuine rate. Kean University's rate came in at roughly seven times the panel's next-highest institution — a real, independent signal alongside its separate, real financial exposure from the NJCU merger, with the honest caveat that its smaller research base makes the rate estimate genuinely noisier than at a major research university.

External Governance Attestation, empirically confirmed (RICD Definition 6.1.1). The King's College's real closure was caused by an accreditor withdrawing accreditation, not financial distress — no continuous feature was ever going to explain it. Tested both as a fitted regression coefficient (resolved King's but introduced a new error elsewhere) and as a direct override per the definition's own design (resolved cleanly, no new error anywhere in the panel) — a genuine empirical confirmation of a design principle previously stated on theoretical grounds alone.

Scale-Normalized Reserve Adequacy, found via a real self-correction (RICD Definition 11.10.5). A peer-density diagnostic (measuring a stalled case's actual distance to its nearest real peers in classifier feature space, not just its broad institutional type) revealed that Limestone and Abilene Christian each sat in a genuinely contested region where a real closure and real stable institutions were nearly indistinguishable. Direct inspection of their real endowment data suggested a missing dimension. An initial fix (endowment normalized by resource debt) looked complete on a smaller subset, then broke once public research universities were included — debt and scale are not independent, so dividing by debt partially cancelled the very variation being controlled for. Normalizing by enrollment scale directly resolved this cleanly on the complete panel, resolving both cases at once. The self-correction is preserved honestly rather than edited out: an early version of this fix was reported as resolved, found incomplete on further testing, and corrected.

A conceptual refinement, not yet tested against data (RICD 10.2). A container's real boundary is constituted by wherever actor energy is actually interfacing with a recognizable pattern — its energetic signature — not by any official claim about that boundary. Ordinary decline is the signature drifting from real activity slowly enough that the gap eventually leaks into a container's own downstream numbers, which is what let the tracker catch real decline years early without observing an actor directly. Fraud, by contrast, is the manufacture of a signature specifically engineered to have no real container behind it. Flagged explicitly as a hypothesis for a future actor-layer tracker to test, not a confirmed finding of this project.

Real, General Methodological Lessons (Feed the Instructional Manual)

Data verification discipline

- Check every variable code against the actual year's own data dictionary before trusting it — meanings and field names do not reliably carry across years (confirmed repeatedly: SAL-format vs. S-series HR files, F2 vs. F1A finance forms, "S[year]_NH" meaning "New Hires" specifically, not general staff headcount).
- A plausible-sounding label is not the same as the correct total — two named sub-categories that both "sound like" spending are not the same as genuine total expenditure.
- "Don't trust the tool is broken; verify labeling and identity first" as a standing default, confirmed across multiple real incidents this session (finance-form misidentification, a data request for data already in hand, institutional ID mismatches).

Model architecture

- Non-centered parameterization is the correct default for any logit/latent random walk — a centered form with a fitted step-size parameter is a textbook sampling trap.
- A parameter with a prior but no likelihood connecting it to a real outcome is decorative, not a real classifier, however plausible it looks in a model's narrative description (the original θ_{hull} diagnosis).
- An accumulator built from a partial slice of real categories, rather than a genuine total, can be structurally unable to ever activate regardless of how severe the real underlying problem is.
- Classify an observable as drift-type or shock-type before choosing how to model it: a quantity with long stable stretches punctuated by rare large jumps needs a heavy-tailed or explicit jump structure, not an ordinary continuous-drift process, which will systematically understate real severity.
- Some diagnostics (in this project, Common-Cause Signature) must run on raw, adapter-level data, not Bayesian model-derived output, since shared-prior smoothing can wash out exactly the sharp signal they are built to detect.

- Not every validated feature belongs inside the same latent-smoothing architecture just because it is technically possible to put it there — a genuinely slow-moving quantity benefits from smoothing between sparse observations; a discrete-event detector does not.

Feature and label validation

- Audit label quality with the same rigor applied to model features — the largest accuracy gain this project produced came from fixing labels, not adding features.
- A well-motivated candidate feature can still make results worse on a small panel; test honestly via held-out validation before trusting a plausible mechanism.
- A feature that fails standalone may still work as an interaction term with a second, related signal.
- Bootstrap-resampling a flagged case's classification (holding it out, refitting on resampled versions of the rest of the panel) distinguishes genuine signal conflict (robustly wrong regardless of training composition) from ordinary small-sample noise (wrong close to a coin flip, expected to resolve with more data) — validated repeatedly and precisely as predicted across this project's real cases.
- When a stalled case resists generic additions, check its actual distance to real peers in the classifier's own feature space, not just its broad institutional type — this can reveal that the real gap is a missing feature dimension, not insufficient data of a similar kind.
- When a case stalls, check whether it shares a specific institutional type with other stalled cases and add matching comparisons rather than generic institutions — validated directly (UNC-Chapel Hill resolved by adding large public research universities specifically, not more small private colleges).
- Some features are ambiguous, not just weak — a raw ratio can genuinely point either direction depending on context not captured by the number itself (confirmed twice: online-enrollment share, and facilities/auxiliary spending relative to instruction), and the fix is the same as for a failed interaction candidate: find a second signal that resolves the ambiguity, or report the feature as a genuine negative result rather than silently dropping it.

Denominators and normalization

- A raw count confounded with scale (in this project, research retractions) is not yet a usable feature until divided by a genuine, comparable denominator.
- Normalize a stock quantity by a genuinely independent scale measure, not by a quantity correlated with scale — dividing reserves by debt looked like a fix on a smaller, homogeneous subset, then broke once institutions spanning a much wider range of absolute scale were included, because debt and scale are not independent. Dividing by enrollment directly resolved it.
- Causal normalization needs a stated, specific rule (expanding range, rolling window, or fixed calibration period) — "use only information available up to now" alone does not determine this, and the choice changes the resulting trajectory.

The four-category baseline (the single most important methodological structure of the project)

Sort every institution into: confirmed healthy (no known confound); confirmed collapse (every real closure, including atypical subtypes — never excluded for looking different from the majority pattern); set aside as false or incomplete (a real feature value measuring something true but not currently relevant, such as stale history or an already-reversed dip); or set aside as confirmed distress or unverified possible collapse (real, uncomfortable signal that is not a measurement artifact). The test for the last two categories is never "does this hurt accuracy" — it is whether the reading is false/incomplete, or true but inconvenient.

Panel representativeness

A validation panel's accuracy is only meaningful if its composition reflects the real population it describes. Deliberately or accidentally over-sampling distinctive, easily-searchable edge cases (a real risk, since dramatic

events generate far more searchable documentation than an ordinary multi-year decline) produces a panel that makes genuine, typical model performance look artificially worse than it is. Confirmed directly: correcting a real 35%-edge-case skew (against a real-world base rate closer to 10-15%) by adding deliberately ordinary closures improved both the honest accuracy figure and resolved an unrelated ambiguous case as a side effect.

Real-world coupling

Prefer a real, currently-identifiable, documented relationship over invented coupling weights wherever one exists — all three real instances in this project (NJCU/Kean, the Atlanta University Center, Dream Center/Argosy) used actual, sourced figures rather than fitted or assumed values. Where the real event postdates the data window or the affected container's own records are too fragmented for a full statistical fit, state the result plainly as a forward projection or a worked instance, not a retrospective validation.